

Homework 1

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Abstract

This is my abstract!

1. QUESTION 1

1.a. Proof

Start from taking logarithm on both sides of return expression:

$$\log R_{e,t+1} = \log(1 + \exp(\log D_{t+1} - \log P_{t+1})) + \log P_{t+1} - \log P_t \quad (1)$$

for simplicity, I denote $r_{e,t+1} = \log R_{e,t+1}$, $p_{t+1} = \log P_{t+1}$, $d_{t+1} = \log D_{t+1}$, and $dp_{t+1} = \log D_{t+1} - \log P_{t+1}$, then by Taylor expansion on dp_{t+1} as it is stationary with mean $\bar{dp}$, we get

$$r_{e,t+1} = \log(1 + \exp(\bar{dp})) + \frac{\exp(\bar{dp})}{1 + \exp(\bar{dp})} (dp_{t+1} - \bar{dp}) + p_{t+1} - p_t. \quad (2)$$

Denote $\kappa = 1/(1 + \exp(\bar{dp}))$, and $\kappa_0 = \log(1 + \exp(\bar{dp})) - \bar{dp}/(1 + \exp(\bar{dp})) = -\log \kappa - (1 - \kappa) \log(1/\kappa - 1)$. The log linearization turns to be

$$r_{e,t+1} = \kappa_0 + (1 - \kappa) dp_{t+1} + p_{t+1} - p_t. \quad (3)$$

Move p_t to LHS and other terms to RHS, we get

$$\begin{aligned} p_t &= \kappa_0 + (1 - \kappa) d_{t+1} - r_{e,t+1} + \kappa p_{t+1} \\ &= \kappa_0 + (1 - \kappa) d_{t+1} - r_{e,t+1} + \kappa [\kappa_0 + (1 - \kappa) d_{t+2} - r_{e,t+2} + \kappa p_{t+2}] \\ &= \kappa_0 (1 + \kappa + \dots + \kappa^{H-1}) \\ &\quad + (1 - \kappa) (d_{t+1} + \kappa d_{t+2} + \dots + \kappa^{H-1} d_{t+H}) \\ &\quad - [r_{e,t+1} + \kappa r_{e,t+1} + \dots + \kappa^{H-1} r_{e,t+H}] \\ &\quad + \kappa^H p_{t+H} \end{aligned} \quad (4)$$

The log dividend price ratio dp_t then can be written as

$$\begin{aligned}
dp_t &= -\kappa_0(1 + \kappa + \dots + \kappa^{H-1}) \\
&\quad + [r_{e,t+1} + \kappa r_{e,t+1} + \dots + \kappa^{H-1}r_{e,t+H}] \\
&\quad d_t - (1 - \kappa)(d_{t+1} + \kappa d_{t+2} + \dots + \kappa^{H-1}d_{t+H}) \\
&\quad - \kappa^H p_{t+H} \\
&= -\kappa_0(1 + \kappa + \dots + \kappa^{H-1}) \\
&\quad + [r_{e,t+1} + \kappa r_{e,t+1} + \dots + \kappa^{H-1}r_{e,t+H}] \\
&\quad d_t - d_{t+1} + \kappa(d_{t+1} - d_{t+2}) + \dots + \kappa^{H-1}(d_{t+H-1} - d_{t+H}) \\
&\quad + \kappa^H d_{t+H} - \kappa^H p_{t+H} \\
&= \kappa_0 \frac{\kappa^H - 1}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H}
\end{aligned} \tag{5}$$

where $\Delta d_{t+1} := d_{t+1} - d_t$. Notice that above identity holds ex post, and it should also hold when taking conditional expectation at time t. Then we get the equation

$$dp_t = \kappa_0 \frac{\kappa^H - 1}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} + \kappa^H \mathbb{E}_t dp_{t+H} \tag{6}$$

Imposing no-bubble condition and let $H \rightarrow \infty$, where

$$\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t dp_{t+H} = 0 \tag{7}$$

, then we have equation (1.3)

$$dp_t = \kappa_0 \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} \tag{8}$$

1.b.

1.c.

1.d. *Alternative log linear present value identity:*

Use same trick

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{D_{t+1}} \frac{D_{t+1}}{D_t} \frac{D_t}{P_t + D_t} \frac{P_t + D_t}{P_t} \tag{9}$$

, log one period return can be written as \$\$

$$r_{e,t+1} = -\log(1 - e^{-dy_{t+1}}) + \Delta d_{t+1} + \log(1 - e^{-dy_t}) + dy_t \tag{10}$$

\$, where $dy_t := \log(1 + \frac{D_t}{P_t}) - \log(1 - e^{-dy_t}) = \log \frac{D_t}{P_t + D_t}$.

Apply Taylor expansion on $\log(1 - e^{-dy_t})$ over dy_t with mean $\bar{dy}$, we have \$\$

$$\log(1 - e^{-dy_t}) = \log(1 - e^{-\bar{dy}}) + \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}} (dy_t - \bar{dy}) + O(dy_t^2) \tag{11}$$

$$Let \tag{12}$$

$$\kappa = e^{-(\bar{dy})}$$

to simplify the expression, (13)

$$\log(1 - e^{-dy_t}) = \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{d}y) + O(dy_t^2) \quad (14)$$

Then the return is loglinearized as (15)

$$r_{e,t+1} = -\frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{d}y) + \Delta d_{t+1} + \frac{\kappa}{1 - \kappa}(dy_t - \bar{d}y) + dy_t \quad (16)$$

and (17)

$$\begin{aligned} \frac{dy_t}{1 - \kappa} &= r_{e,t+1} - \Delta d_{t+1} + \kappa \frac{dy_{t+1}}{1 - \kappa} \\ &= r_{e,t+1} - \Delta d_{t+1} \\ &\quad + \kappa \left[r_{e,t+2} - \Delta d_{t+2} + \kappa \frac{dy_{t+2}}{1 - \kappa} \right] \\ &= r_{e,t+1} - \Delta d_{t+1} + \kappa [r_{e,t+2} - \Delta d_{t+2}] \end{aligned} \quad (18)$$

$$\begin{aligned} &\quad + \dots + \kappa^{H-1} [r_{e,t+H} - \Delta d_{t+H}] + \kappa^H \frac{dy_{t+H}}{1 - \kappa} \\ &= \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H \frac{dy_{t+H}}{1 - \kappa} \end{aligned} \quad (19)$$

Thus, the loglinearization identity is (19)

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H} \quad (20)$$

Take conditional expectation at time (21)

t, identity also holds

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} \right) + \kappa^H \mathbb{E}_t dy_{t+H} \quad (22)$$

When imposing no-bubble assumption, the last term vanishes with (23)

h arrow r infinity,

$$\lim_{h \rightarrow \infty} \kappa^H E_t \log \left(\frac{P_t + D_t}{P_t} \right) < \kappa^H E_t \left[1 + \log \left(\frac{D_t}{P_t} \right) \right] \rightarrow 0 \quad (24)$$

2. QUESTION 2